

Quiz 1 - Computational Physics I

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SCORE: 16/20

Date: Wednesday 26 February 2025 Duration: 45 minutes

Credits: 20 points (5 questions) Type of evaluation: LAB

Provide short and concise answers to the following items:

1. (4 points) Programming Languages

(a) Explain the difference between low-level and high-level programming languages.

(b) Provide 1 example of a low-level programming language and 1 of a high-level language.

a) ~~Low~~ ^{High}-level languages are user-friendlier and easy to understand, contrary to low-level languages. This is because of its architecture, HLL, are several steps larger to interpret and compile by the machine, while LLL are close to the computer's language and therefore faster. b) L.L.L.: C, Fortran H.L.L. Python.

2. (4 points) Python Functions

(a) Explain what a Python function is.

(b) List 4 types of Python functions.

a) A function is a preconfigured series of steps that can be applied to the correct ^{block of code} input, giving our desired output depending on the function workflow. How?

b) Recurses ^{For is control statement.}
↓
for ...
descriptive [↓] len()
quantitative [↓] max(x)
qualitative [↓] type()

3. (4 points) Lambda Functions and Vectorisation

(a) Briefly explain what a lambda function is in Python.

(b) Briefly explain the concept of vectorisation in Python.

a) lambda functions are a faster way to create a mathematical function, which ~~lambdifies~~ ^{lambdifies} the selected variable and follows the steps, mathematical, assigned to it. How? 1-line single expression. Implicit return.

b) Python is a vectorised language because the functions and methods can handle single values or whole vectors without needing to specify or change, correctly handling both or N-d arrays.

4. (4 points) Linux File Permissions

In a Linux environment, the file permissions for a script named `run_test.sh` are currently set as `-rw-rw-r--`. You need to allow all users to execute the script but want to keep the current read and write permissions unchanged.

- Explain the meaning of the current file permissions `-rw-rw-r--` for `run_test.sh`.
- Provide the `chmod` command you would use to modify the permissions so that all users can execute the script without altering the read and write permissions.

-1.3

- After applying the `chmod` command, what will the new permissions of the file `run_test.sh` be? Write them in both symbolic and numeric forms.

a) $r = \text{read}$ ✓
 $w = \text{write}$ ✓
 $\ominus$ `-rw-rw-r--`
 What about this character?
 ↓
 user permission alongside users permission public permissions ✓

b) `sudo chmod 775 /run_test.sh` ✓

c) `-rwx-rwx-r-x-` ✓
 # is are the other way around
 $\begin{matrix} 1 & 2 & 4 \\ \times & \times & \times \end{matrix}$
 $\begin{matrix} 7 & 7 & 5 \end{matrix}$

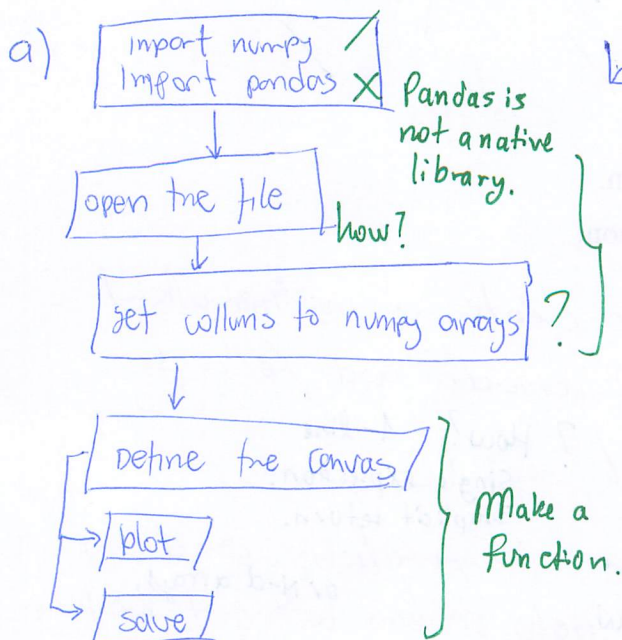
$r \rightarrow 4$
 $w \rightarrow 2$
 $x \rightarrow 1$

5. (4 points) Python I/O and Plotting

In an optics experiment, a laser beam is passed through a single-slit aperture, and the resulting diffraction pattern is recorded on a screen. A photodetector scans across the screen, measuring the light intensity at different positions. The recorded data is stored in a text file named `diffraction_data.txt`, which contains two columns: the first column represents the position on the screen (x); the second column represents the measured light intensity (I).

-1.7

- Sketch an algorithm workflow to read the data from the file, and generate and save a plot of light intensity vs. position using Python.
- Write the overall syntax of a native I/O function that reads the data from a given file, and returns the extracted x and I values as NumPy arrays.



b) `def .....`
`data = open("~/diffraction_data.txt", "r")` ✓
 Open empty list.
`for i in data:`
`print(i)` → reader is missing.
 Make an I/O function
 ↳ set values as float
 ↳ append → this is for lists.
 ↳ to np.array. X